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In-depth insights into molecular mechanisms of enhanced boiling heat transfer through interpretable machine learning

Hao Xiao [a,b], Tao Yang [a,b], Yi Zheng [a,b,*], Guangwen Zhang [a,b], Ming Ma [a,b], Chunyu Jiang [a,b], Chun Wang [a,b], Xiangdi Zhao [a,b,*]

[a] State Key Laboratory of Chemical Safety, Qingdao 266000, PR China
[b] SINOPEC Research Institute of Safety Engineering Co., Ltd., Qingdao 266000, PR China

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ABSTRACT

Understanding the molecular mechanisms underlying boiling heat transfer and accurately predicting additive performance are crucial for the rational design of surfactants that enhance thermal efficiency and safety. In this study, interpretable machine learning models were developed to predict the heat transfer coefficient (HTC) and critical heat flux (CHF), while revealing the influence of surfactant structural features on these properties. Two neural network architectures combined with ten optimization algorithms were systematically evaluated, and the feed-forward back-propagation network using the Levenberg–Marquardt algorithm exhibited the best performance. Based on conventional molecular descriptors, the model achieved a correlation coefficient of 0.9306, confirming its predictive reliability. Interpretable analysis indicated that molecular polarity predominantly governs variations in HTC and CHF. To capture more nuanced polarity features, descriptors derived from the conductor-like screening model were introduced, improving the correlation coefficient to 0.9797. Feature-importance evaluation revealed that charge density accounted for 86 % of the variance in HTC and CHF. Sensitivity analysis further showed that non-polar regions strongly promoted HTC but reduced CHF, while hydrogen-bond acceptor regions exhibited the opposite effect. Mechanistic analysis suggested that non-polar regions facilitate bubble nucleation yet accelerate vapor film formation, whereas hydrogen-bond acceptor regions stabilize the liquid microlayer and delay film boiling. These findings demonstrate that molecular structure and charge density fundamentally dictate surfactant-influenced boiling performance, providing a data-informed and theory-driven framework for rational surfactant design toward safer and more efficient thermal management systems.

1. Introduction

Boiling heat transfer is recognized as an extremely efficient phase-change mechanism [1,2]. It plays an indispensable role in numerous applications, including the energy and chemical industries, thermal management of electronic devices, aerospace engineering, and nuclear energy systems [3–5]. Compared to single-phase convection, boiling achieves significantly higher heat transfer rates by leveraging two-phase vapor-liquid flows and phase change phenomena [4]. However, as technological and energy demands increase, conventional boiling systems often fail to meet performance requirements under extreme operating conditions [6]. To address this challenge, heat transfer enhancement technology focus on optimizing two critical parameters: the heat transfer coefficient (HTC) and the critical heat flux (CHF) [7,8]. HTC quantifies the efficiency of heat exchange between a solid surface and the working fluid, representing the rate of heat transfer per unit area and per unit temperature difference [8]. CHF is the maximum heat flux in nucleate boiling; beyond this point, the surface transitions to a less efficient boiling regime [9]. Exceeding the CHF triggers a sharp rise in solid surface temperature, a phenomenon known as the *"boiling crisis"* [10].

In recent years, research on boiling heat transfer enhancement has concentrated on two primary strategies: surface modification and fluid modification [11–15]. In surface modification, micro-structured surfaces have achieved significant gains in HTC and CHF [11]. However, this approach faces challenges such as complex fabrication processes, mechanical stability issues, and surface degradation over time [16]. In fluid modification, nanofluids have been shown to increase CHF

* Corresponding authors at: State Key Laboratory of Chemical Safety, Qingdao 266000, PR China.
E-mail addresses: liuwujushi.123@163.com (Y. Zheng), zhaoxd.qday@sinopec.com (X. Zhao).

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[12,17]; however, nanoparticle agglomeration and sedimentation can degrade heat transfer performance [18,19], and high nanofluid concentrations markedly increase flow resistance. In contrast, adding surfactants can enhance boiling performance without altering equipment, offering practical advantages such as adjustable concentration, low cost, and ease of use.

Most surfactants tend to significantly increase HTC while simultaneously reducing CHF [20,21]. To develop surfactants that improve both HTC and CHF, it is crucial to understand how molecular structure influences heat transfer performance [21–23]. For example, Wen et al. [21] examined the effect of functional ionic groups in surfactant molecules on HTC and CHF, finding that surfactants with identical or similar ionic groups exhibit similar boiling performance. Luo et al. [22] found that a rigid steroidal backbone in surfactant molecules promotes the formation of nucleation sites, thereby increasing HTC. However, the diversity of surfactant structures and their complex interactions with boiling phenomena mean that experiments alone are costly and insufficient for uncovering the fundamental mechanisms of heat transfer enhancement.

Establishing predictive models of HTC and CHF based on key parameters enables strategic regulation of experimental variables—focusing on the most influential parameters and thus drastically reducing the number and cost of necessary trials [24,25]. For instance, Kinjo et al. [26] established a model to predict the effects of physical properties such as saturation pressure on HTC. Li et al. [27] built a mechanical model within the Euler–Lagrange framework to evaluate the boiling behaviors. Nevertheless, the existing studies often focus on bulk fluid properties (e.g., concentration, viscosity) while neglecting molecular-scale features that surfactant structure [24,26,27]. However, evidence shows that surfactant solutions with identical fluid properties but differing functional groups can exhibit different heat-transfer performance [16]. Hence, it is essential to employ molecular-level modeling to capture these effects. Recent advances in quantitative structure-property relationship (QSPR) model based on the machine learning (ML) have enabled the linking of molecular structure to macroscopic material properties [28–32]. Javed et al. [33] demonstrated that neural networks are highly effective machine-learning models, producing predictions that are in excellent agreement with experimental data. Zhang et al. [31] found that the reasonable selection of molecular descriptors is the key to predicting performance and revealing mechanisms. However, the research on the QSPR model of surfactant enhanced boiling heat transfer is still insufficient and there is no systematic framework to integrate molecular descriptors and boiling heat transfer data into surfactant design.

In summary, the rational design of surfactants to enhance thermal efficiency and safety is of paramount importance. However, the vast diversity of molecular structures and their intricate interactions with boiling phenomena render purely experimental investigations cost-prohibitive. This challenge underscores the need for predictive modeling strategies. Although prior studies have linked bulk fluid properties to boiling performance, they frequently overlook the molecular-scale features of surfactant structures. To address this gap, multiple molecular descriptors were integrated with neural-network architectures to predict surfactant performance and unravel the molecular mechanisms driving boiling enhancement. Firstly, cascade-forward back-propagation (CFBP) neural networks and feed-forward back-propagation (FFBP) neural networks were used under 10 algorithms to evaluate model and algorithm performance. Subsequently, the FFBP model with FAST descriptors that encompass structural, thermodynamic, and topological characteristics was developed and the sensitivity analysis of the inputs to the outputs were carried out. Then, in view of these analysis, a model with cavity volume and σ-profile descriptors was established to further characterize the structure-property relationship quantitatively. This work elucidates the mechanisms by which molecular characteristics regulate boiling enhancement, thereby providing theoretical guidance for the development of novel surfactant-enhanced heat-transfer systems.

2. Methods

2.1. Acquiring experimental data sets

A total of 138 experimental data points of surfactant-enhanced pool boiling on smooth heating surface were obtained from the literature, covering 21 surfactants at various concentrations [20–23,34–37]. To reduce variability across studies, the reported HTC and CHF values for each surfactant were normalized by dividing them by the corresponding values for pure water under comparable conditions. This yielded intensification factors for HTC and CHF, which were used as target variables in model development.

2.2. Molecular descriptors

Two classes of molecular descriptors were computed for each surfactant. The first type is the FAST descriptor, which is based on structural, thermodynamic, and topological characteristics. It includes the number of hydrogen bond donor (No. HBD), the number of hydrogen bond acceptor (No. HBA), molecular refractivity (MR), ALogP, Balaban index (BI), Wiener index (WI), complementary and structural information contents (CIS and SIC). Another class of descriptors is the conductor-like screening model (COSMO) descriptors. COSMO is commonly used to predict the thermodynamic properties of compounds by creating a virtual conductor around each molecule and calculating the surface area and screening charge density of the surface segments of molecule to obtain the σ-profile and cavity volume [38–40].

Descriptor generation was performed using Materials Studio as follows. Surfactant structures were first geometrically optimized in the DMOL3 module with the GGA/VWN-BP functional. FAST descriptors were then extracted via the FAST engine in QSPR module.

After energy calculations on optimized structures, COSMO-derived features (cavity volume and σ-profile) were obtained. To reduce dimensionality, σ-profile regions were integrated into six aggregated areas, exemplified for dodecyltrimethyl-ammonium bromide (DTAB) in Fig. 1. The regions are defined as:

S1: $-0.03\ e/A^2 \le \sigma \le -0.02\ e/A^2$ (strong HBD region).
S2: $-0.02\ e/A^2 < \sigma \le -0.0082\ e/A^2$ (weak HBD region).
S3: $-0.0082\ e/A^2 < \sigma \le 0\ e/A^2$ (non-polar region).
S4: $0\ e/A^2 < \sigma \le 0.0082\ e/A^2$ (non-polar region).
S5: $0.0082\ e/A^2 < \sigma \le 0.02\ e/A^2$ (weak HBA region).

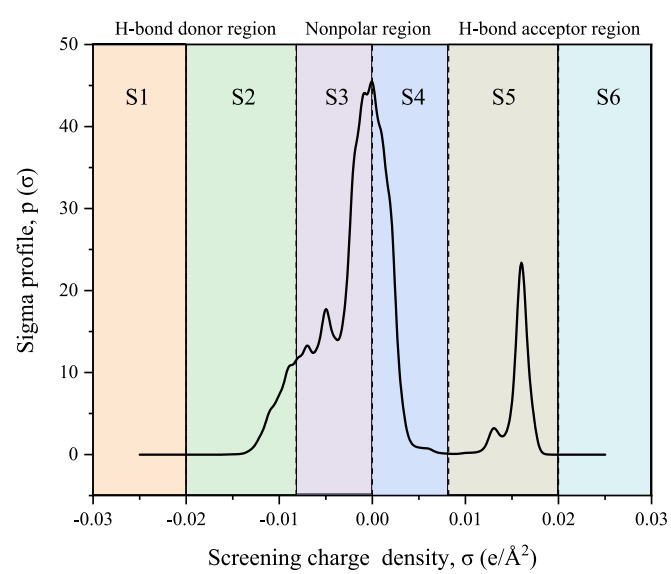

Fig. 1. Representation of the six S_σ descriptors for the DTAB.

S6: 0.02 e/A^2 < σ ≤ 0.03 e/A^2 (strong HBA region).

2.3. Development of the machine learning model

In this study, the FFBP and CFBP neural network architectures were selected due to their well-established theoretical foundations, structural simplicity, and suitability for modeling nonlinear relationships. In both configurations, the error at the output layer is propagated backward through successive neuron layers, with weights and biases adjusted iteratively to minimize the overall loss. In addition, these two architectures allow for a controlled comparison of network topology under consistent training conditions. Moreover, both models are relatively easy to implement and interpret, offering clear connections between inputs, hidden layers, and outputs. As illustrated in Fig. 2, FFBP neural network is a multi-layer feed-forward neural network trained according to the error back-propagation algorithm and is one of the most widely used neural network models. It calculates the gradient of the loss function with respect to the weights by propagating errors backward from the output to adjust the weights. In this study, a three-layer architecture was adopted: an input layer for molecular descriptors, one hidden layer with 15 neurons, and an output layer for predicted intensification factors. The input layer receives the aforementioned descriptors (such as FAST descriptors consist of 9 features). The hidden layer applies weight matrices and bias vectors to transform this input before forwarding it to the output layer. For each input vector i, the hidden-layer activation $f_1(a_1)$ is calculated using Eq. (1), and the output-layer activation $f_2(a_2)$, corresponding to HTC and CHF predictions, is obtained from Eq. (2). Here, w and b denote the weight matrices and bias vectors, respectively, and f represents the activation function as defined in Eqs. (3) and (4).

$$f_1(a_1) = f_1(w_1 \times i + b_1) \tag{1}$$

$$f_2(a_2) = f_2(w_2 \times f_1(a_1) + b_2) \tag{2}$$

$$f_1(x) = 2/\left(1 + e^{-2x}\right) - 1 \tag{3}$$

$$f_2(x) = x \tag{4}$$

CFBP extends the standard FFBP topology by adding direct connections from the input layer to every hidden layer as shown in Fig. 3. This "cascading" of the original input into deeper layers creates multiple parallel paths for both forward signals and backward gradients, which can accelerate convergence and help alleviate vanishing-gradient issues. The output-layer activation, $f_2(a_2)$, is determined by Eq. (5) in CFBP neural network model.

$$f_2(a_2) = f_2(w_2 \times f_1(a_1) + b_2 + w_0 \times i) \tag{5}$$

Subsequently, w and b were solved by minimizing the sum of the errors between the predicted and target values using the training algorithm. In this study, ten training algorithms were implemented for neural network optimization and grouped by their update mechanisms. First-order methods include gradient descent (GD), gradient descent with momentum (GDM), gradient descent with adaptive learning rate (GDA), and gradient descent with momentum and adaptive learning rate (GDX). Conjugate-gradient methods comprise conjugate gradient back-propagation (CGB) and scaled conjugate gradient (SCG), while quasi-Newton methods consist of Broyden-Fletcher-Goldfarb-Shanno (BFG) and Newton-Raphson (NR). The one-step secant method (OSS) and the Levenberg-Marquardt algorithm (LM) were also evaluated. All algorithms were applied to both FFBP and CFBP architectures.

In addition, the dataset was randomly partitioned into training (70 %), validation (15 %), and testing (15 %) subsets. Early stopping was guided by validation performance, and final evaluation was conducted on the independent test set.

2.4. Model validation, performance and analysis

To assess predictive performance, the correlation coefficient (R), mean squared error (MSE), and the values of error were computed according to Eqs. (6–8), where N denotes the number of samples, and t and o represent the true target and model output, respectively.

$$MSE = \frac{1}{N}\sum_i |t_i - o_i|^2 \tag{6}$$

$$R = \frac{N\left(\sum_i t_i o_i\right) - \left(\sum_i t_i\right)\left(\sum_i o_i\right)}{\sqrt{\left[N\sum_i t_i^2 - \left(\sum_i t_i\right)^2\right]\left[N\sum_i o_i^2 - \left(\sum_i o_i\right)^2\right]}} \tag{7}$$

$$\text{Error} = t_i - o_i \tag{8}$$

In addition, in order to explore the effect of different inputs on the outputs, sensitivity and importance analyses were performed via partial derivatives. For each training sample, the first derivative of each network output with respect to each input descriptor was calculated. These derivatives were averaged over all samples to obtain sensitivity metrics (mean values of partial derivative) and importance metrics (root mean square values, RMS), quantifying how variations in each molecular descriptor influence model outputs. The partial derivative of the p-th output with respect to the i-th input, evaluated through neuron k, is represented by the Jacobian element $J_{p,i}(x)$, as defined in Eq. (9). For a dataset containing N samples, the corresponding mean and RMS values across all samples are computed as shown in Eqs. (10) and (11), respectively.

$$J_{p,i}(x) = f_2'(a_{2p}) \sum_{k=1}^{15} w_{2p,k} f_1'(a_{1k}) w_{1k,i} \tag{9}$$

$$Mean_{p,i} = \frac{1}{N}\sum_{n=1}^{N} J_{p,i}{}^{(n)} \tag{10}$$

$$RMS_{p,i} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(J_{p,i}{}^{(n)}\right)^2} \tag{11}$$

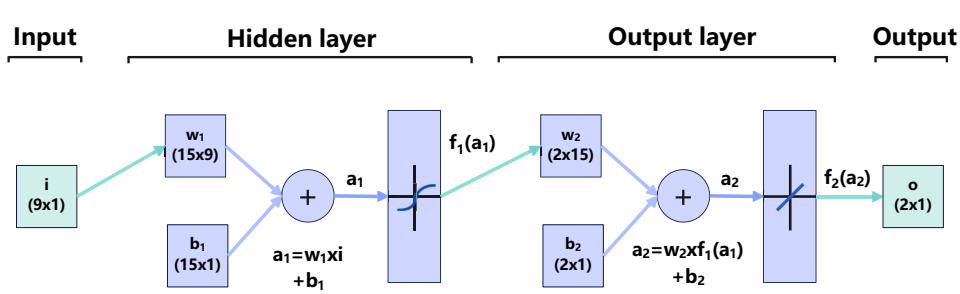

Fig. 2. Schematic diagram of FFBP neural network model.

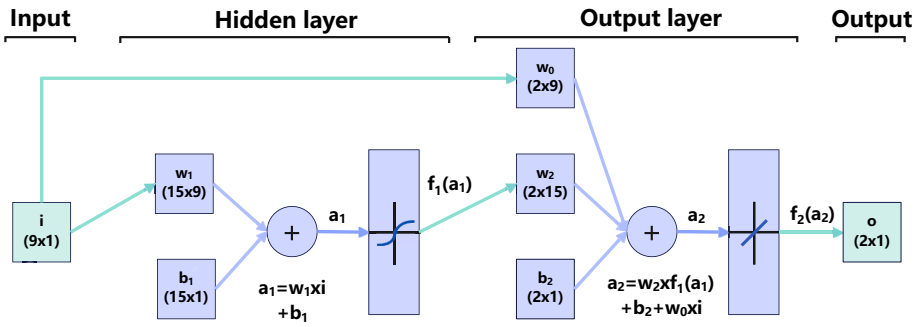

Fig. 3. Schematic diagram of CFBP neural network model.

3. Results and discussion

3.1. Comparison of model architectures and algorithms

A comparison of the *MSE* obtained from two network architectures, FFBP and CFBP, across ten training algorithms is presented in Fig. 4. Overall, the FFBP model outperformed the CFBP model for the majority of algorithms tested. Although the cascade connections in CFBP allow the raw input features to be propagated directly into deeper layers, thereby enhancing representational richness, this complexity appears to compromise generalization. Under the small-sample conditions of this study, the simpler FFBP architecture proved more robust, delivering greater stability in model performance.

Moreover, each of the ten algorithms demonstrated unique convergence characteristics and accuracy profiles. The LM algorithm achieved the lowest MSE, reflecting its superior convergence properties and high fitting precision. This performance stems from its hybrid optimization strategy that interpolates between the Gauss-Newton method and gradient descent, dynamically adjusting the damping factor to leverage both first-order gradients and second-order curvature information. This enables the algorithm to make larger, more informed parameter updates in regions of smooth curvature, while retaining stability in ill-conditioned areas. SCG followed closely behind, demonstrating an excellent balance between convergence speed and numerical stability without the need for explicit line searches. In addition, first-order methods enhanced with momentum and adaptive learning rates, such as GDX and GDA, also surpassed standard gradient descent, underscoring the benefits of these modifications in first-order optimization. In light of these findings, the FFBP network trained with the LM algorithm was selected for all subsequent modeling tasks in this study.

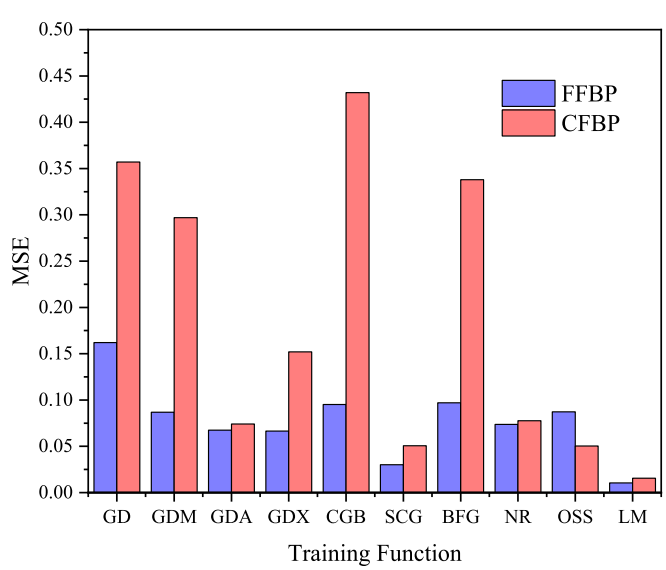

Fig. 4. MSE comparison under different models.

3.2. FAST descriptors

3.2.1. Model validation and performance

Fig. 5 compares the experimental and predicted values of HTC and CHF for the model using surfactant concentration and eight FAST molecular descriptors. The results include a training set (98 data points), a validation set (20 points), and a test set (20 points). It is clear that the training, validation, and test data all lie closely along the diagonal, with only a few outliers likely due to experimental measurement biases. The correlation coefficients (*R* values) for the training, validation, and test sets were 0.9336, 0.9122, and 0.9595, respectively. The overall MSE and R values for the entire dataset were 0.0105 and 0.9306, respectively. These metrics indicate that the predictive performance of the model is generally high. Fig. 6 shows histograms of the deviations between predicted and experimental values of HTC and CHF, with most errors falling within ±0.6.

3.2.2. Model analysis

Fig. 7 shows the importance of input variables for predicting CHF and HTC. Concentration has the highest importance for both HTC and CHF, indicating that this parameter strongly influences boiling heat transfer. HBA, HBD, and ALogP also have high importance, followed by topological descriptors such as the Wiener index.

For molecular descriptors, the ALogP descriptor quantifies lipophilicity: higher ALogP means a more non-polar molecule (lower ALogP means more polar). HBD and HBA count the hydrogen-bond donors and acceptors, characterizing molecular polarity. Topological descriptors (e.g. Wiener index, Balaban index, information content metrics) capture aspects of molecular structural complexity. These results suggest that molecular polarity (HBD/HBA and ALogP) and structural complexity strongly influence HTC and CHF.

In addition, this study used partial derivative mean values to describe the sensitivity analysis of the inputs to the outputs and to interpret the model. A positive sensitivity indicates that increasing the input raises the output, while a negative sensitivity indicates the opposite. For HTC (Fig. 8), increasing surfactant concentration, ALogP, and HBA favorably increase HTC, whereas increasing Wiener index or complementary information content have a negative effect on HTC. In contrast, as can be seen from Fig. 9, increasing concentration, ALogP, or Wiener index tends to reduce CHF, whereas increasing HBD or HBA tends to increase CHF.

As surfactant concentration increases, more molecules adsorb at the liquid-vapor and solid-liquid interfaces, reducing surface tension. Surface tension governs the energetic cost for creating a curved vapor-liquid interface via the Laplace relation: higher surface tension imposes greater pressure on a small-radius vapor embryo, raising the nucleation energy barrier. Surfactant-induced tension reduction lowers this barrier and decreases the critical nucleation radius; consequently, at a fixed wall superheat, more nucleation sites activate and the bubble birth rate rises,

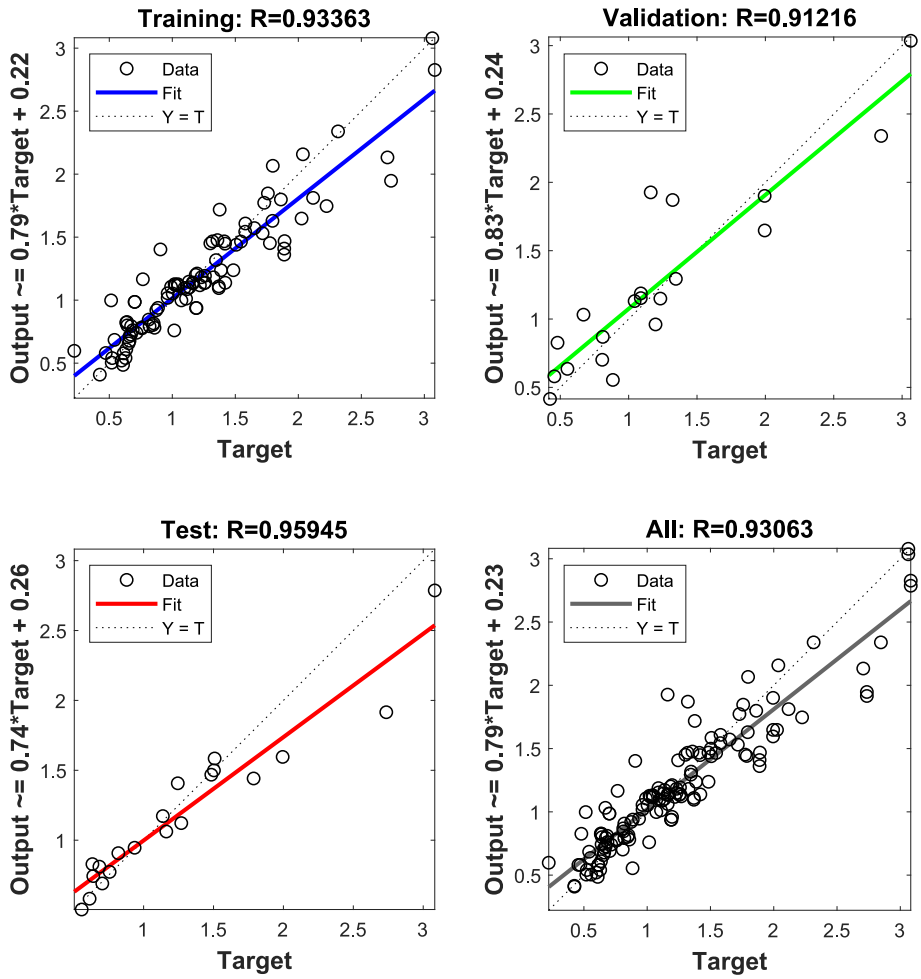

Fig. 5. Regression plots of the model with FAST descriptors.

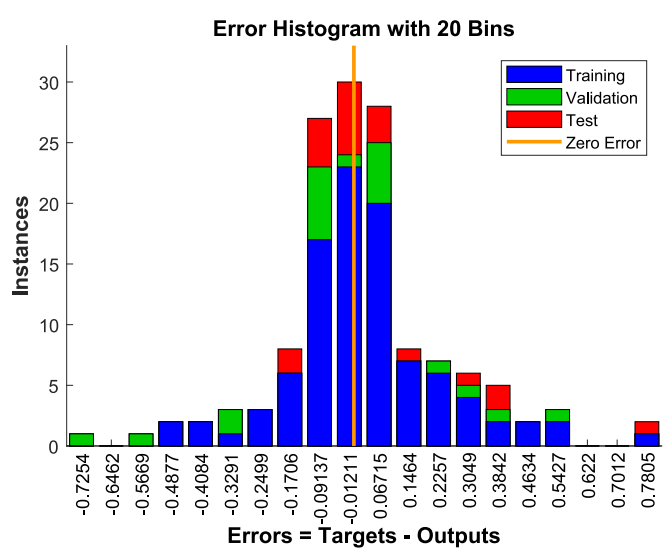

Fig. 6. Error histogram of the model with FAST descriptors.

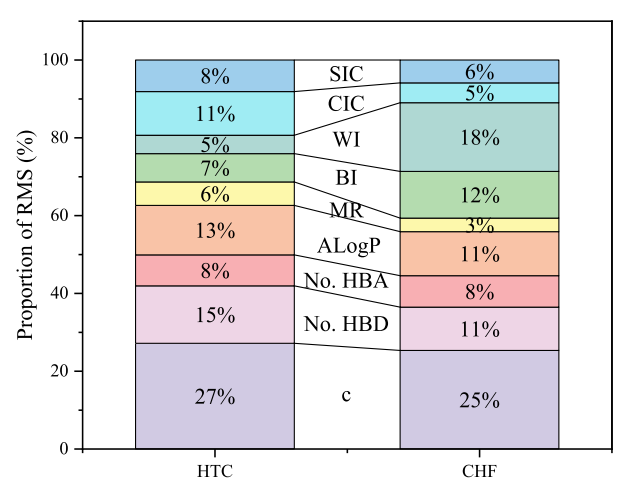

Fig. 7. Importance of the FAST descriptors for CHF and HTC.

explaining the observed increase in HTC. CHF depends on the balance between vapor generation and liquid replenishment at the heated surface. Reduced surface tension diminishes capillary pressure and impairs liquid-film rewetting, making the film prone to instability. Simultaneously, accelerated bubble production favors vapor accumulation. When vapor production outpaces liquid replenishment, coalescing bubbles form a continuous vapor film that disrupts surface wetting and triggers a sharp CHF decline.

Molecular structure further regulates gas-liquid-solid interfacial interactions through synergistic mechanisms. Polar functional regions (HBA/HBD) strengthen hydrogen bonding with interfacial water, enhancing local cohesion, stabilizing the liquid microlayer and delaying dry-out; this raises the nucleation barrier but increases CHF. In contrast,

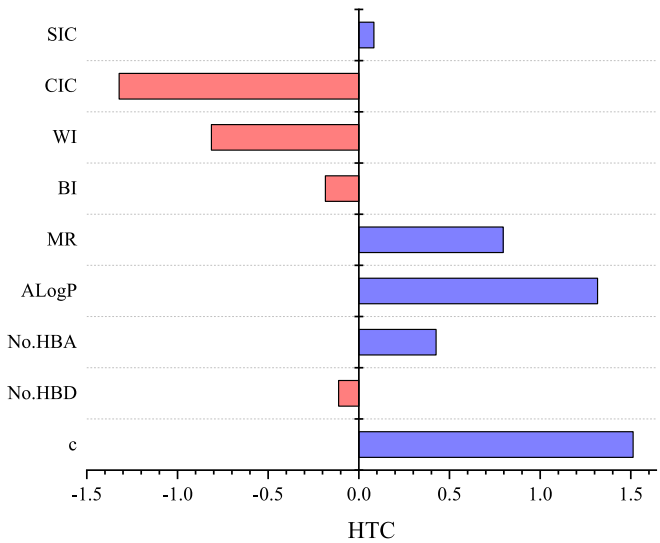

Fig. 8. Sensitivity analysis of the FAST descriptors to the HTC.

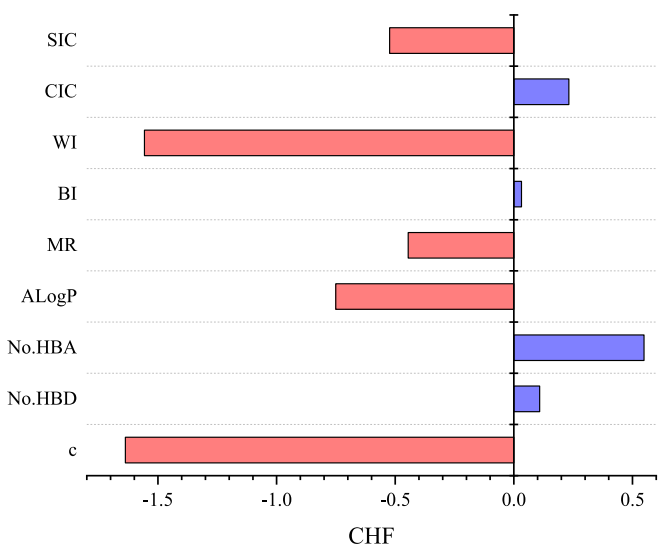

Fig. 9. Sensitivity analysis of the FAST descriptors to the CHF.

non-polar fragments (hydrophobic moieties, alkyl chains) lower local surface energy, facilitate vapor embryo formation and accelerate bubble detachment; these actions boost HTC but undermine liquid-film stability, reducing CHF. Moreover, the topological descriptors such as the Wiener index reflect the molecular size, branching, and spatial distribution of atoms, which influence solubility, interfacial packing, and diffusivity. More complex structures may form more stable interfacial layers or interact differently with the bulk fluid, thereby altering nucleation behavior and vapor film dynamics.

In summary, molecular polarity (HBD, HBA and ALogP), along with structural complexity, strongly influence HTC and CHF. However, FAST descriptors may inadequately capture these molecular characteristics. To improve the QSPR modeling, COSMO descriptors (cavity volume and σ-profile) were used in the next model.

3.3. COSMO descriptors

3.3.1. Model validation and performance

Fig. 10 compares experimental and predicted HTC and CHF values for the model using surfactant concentration and COSMO descriptors. The training, validation, and test points all lie close to the diagonal line, with only a few outliers. The correlation coefficients for the training, validation, and test sets were 0.9921, 0.9642, and 0.9415, respectively, indicating strong performance. Fig. 11 shows the distribution of residuals (predicted minus experimental values) for HTC and CHF. Most residuals were near zero and the largest deviation (~0.859) likely arise from experimental uncertainty. Compared to the FAST-descriptor model, the COSMO-based model achieves higher performance, with an overall correlation coefficient $R = 0.9797$ and an *MSE* of 0.0094.

3.3.2. Model analysis

Fig. 12 illustrates the hierarchical importance of input variables for predicting HTC and CHF in the COSMO descriptor-based model. For HTC, descriptors S5 and S6 are most important, followed by concentration and descriptors S2-S4; cavity volume (V) and S1 have minimal influence. For CHF, descriptors S4-S6 dominate, followed by concentration and S2-S3, while V and S1 again have negligible impact.

For HTC (Fig. 13), increasing the concentration and the non-polar region S4 significantly enhances the HTC. Regions S1 and S2, representing strong and weak HBD areas, respectively, exert a mild positive influence. S4 corresponds to weakly electrostatic, hydrophobic fragments, often alkyl chains or other low-polarity moieties, while S1 and S2 map to hydrophilic sites. These amphiphilic features disrupt local water structure at the interface and lower the energetic cost of vapor embryo formation, thereby raising nucleation density and accelerating bubble birth-departure cycles. The net effect is enhanced microlayer evaporation and convective heat transfer at the heated wall. In contrast, HBA-rich regions (S5, S6), typically carbonyls, ethers, or halogenated groups, stabilize hydrogen-bond networks with interfacial water, strengthening local cohesion. While this improves wettability, it raises the nucleation energy barrier and suppresses boiling dynamics, leading to reduced convective and phase-change heat transfer. In addition, the cavity volume shows a slight negative effect, likely due to hindered diffusivity and diminished interfacial activity, which impair thermal coupling between the surface and bulk fluid.

For CHF (Fig. 14), a nearly inverse trend emerges, reflecting the classic trade-off between bubble generation and liquid-film stability. Polar HBA-rich regions anchor liquid to the surface via directional hydrogen bonding and enhanced polar interactions, stabilizing the thin liquid film and enhancing capillary-driven and molecular-scale rewetting under high heat flux, thereby increasing CHF. HBD-rich regions exhibit a dual effect: they improve surface wettability and rewetting—boosting CHF—yet can also accelerate vapor nucleation (and thus vapor production) under certain conditions; when nucleation acceleration prevails, CHF declines. The observed net negative impact on CHF indicates that, for the studied surfactants, nucleation enhancement outweighs the rewetting benefit. Likewise, greater coverage of non-polar regions (S4) or high surfactant concentration accelerates vapor production and bubble coalescence, fostering early vapor-film formation and impairing liquid-film regeneration, and triggering premature transition to film boiling.

These results highlight the importance of molecular-scale interfacial charge distribution in determining boiling heat transfer behavior. Non-polar regions (e.g., S4) facilitate HTC through nucleation enhancement, while polar HBA regions (S5, S6) support CHF via stabilization of the interfacial liquid layer. The dual role of HBD regions reflects their ambivalent nature in modulating both nucleation and rewetting. In addition, it is worth noting that HBA and HBD show slightly different effects in the FAST and COSMO models. This is likely because the FAST descriptor counts are based on simple structural definitions, whereas COSMO captures the actual charge distribution. For example, some molecules have HBD = 0 in FAST descriptors but exhibit significant positive HBD regions in the COSMO descriptors.

In summary, optimizing surfactant-enhanced boiling requires navigating an intrinsic trade-off between CHF and HTC. In efficiency-driven applications—such as high-performance electronic cooling (chips, servers, laser devices), compact micro-heat exchangers in avionics and

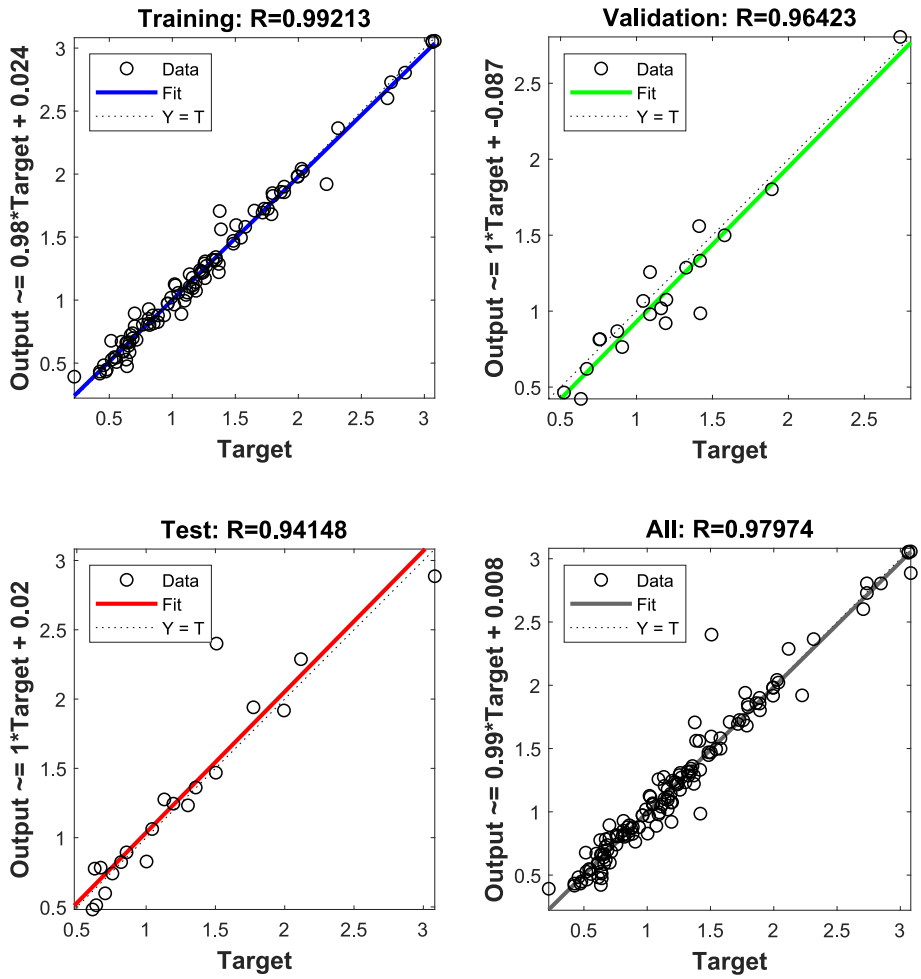

Fig. 10. Regression plots of the model with COSMO descriptors.

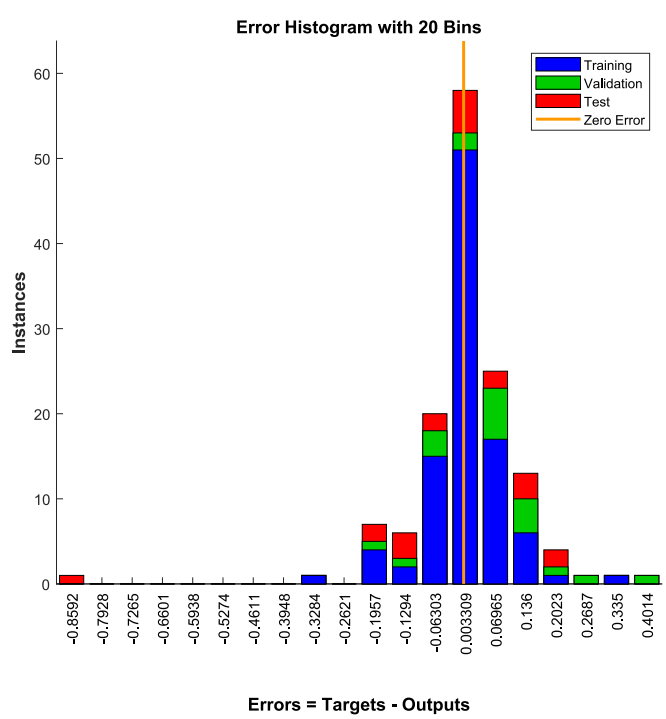

Fig. 11. Error histogram of the model with COSMO descriptors.

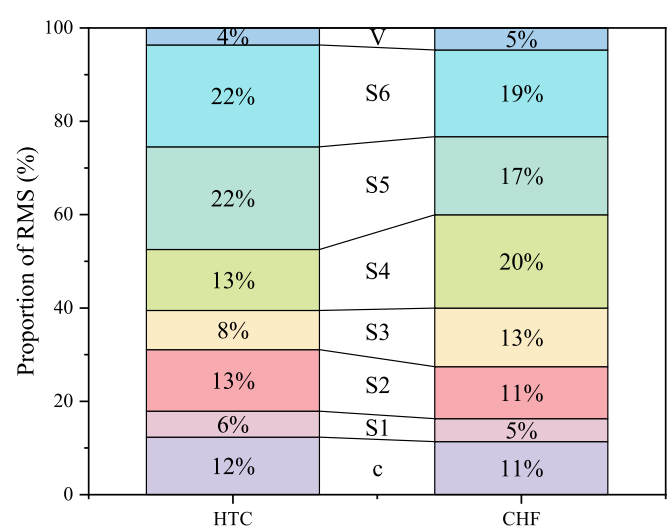

Fig. 12. Importance of the COSMO descriptors for CHF and HTC.

medical instruments, and systems exposed to transient high heat fluxes—maximizing HTC prevents local overheating by accelerating heat removal. In these contexts, surfactant formulations with elevated concentrations and extensive non-polar domains promote bubble nucleation and intensify boiling, thereby boosting heat transfer. Careful concentration control is required, however, to avoid premature film-boiling

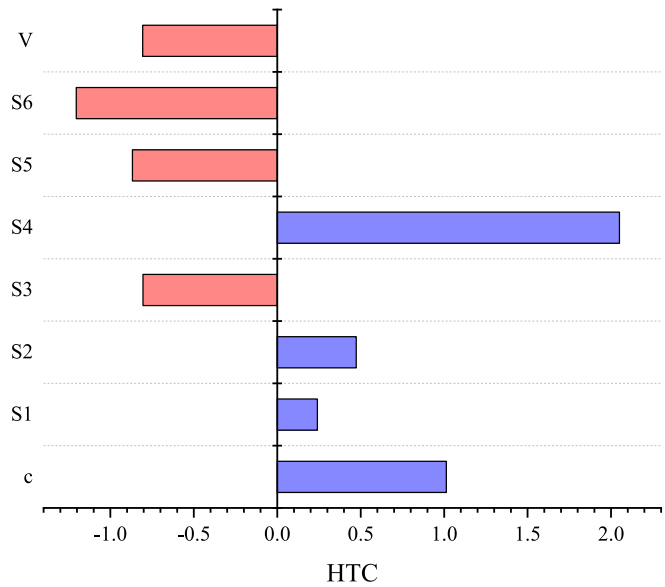

Fig. 13. Sensitivity analysis of the COSMO descriptors to the HTC.

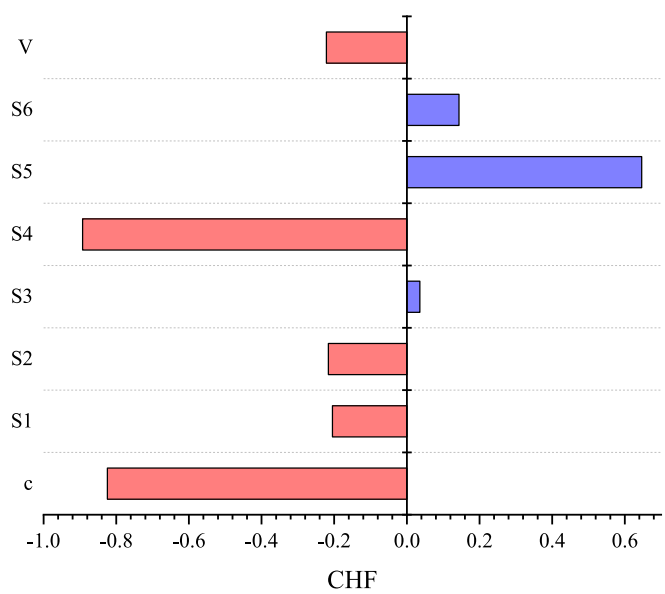

Fig. 14. Sensitivity analysis of the COSMO descriptors to the CHF.

transitions that compromise thermal stability. By contrast, safety- and stability-critical applications—such as nuclear reactor cooling, high-power electronics in electric vehicles and radar systems, and emergency fire-suppression of petrochemical storage tanks—demand enhanced CHF. Low-concentration formulations incorporating polar molecules with multiple HBA regions delay vapor-film formation, sustain liquid–surface contact at elevated heat fluxes, and ensure reliable operation under extreme conditions. In fact, the relationships between molecular descriptors and boiling behavior are fundamentally nonlinear: many features display nonmonotonic effects whose direction and magnitude depend on additive concentration, the spatial arrangement of polar versus non-polar regions, and overall cavity volume. Crucially, strategies that maximize HTC often compromise CHF, necessitating an application-specific balance between heat-transfer enhancement and boiling stability. These observations underscore the complex, nonlinear structure–property interdependencies that must be managed when designing and optimizing surfactant additives for boiling heat transfer.

4. Conclusions

In this study, ML models based on molecular descriptors were developed to elucidate how surfactant molecular properties govern HTC and CHF. The performances of two network architectures across ten training algorithms were compared, and the FFBP network trained with LM algorithm was selected. Using FAST descriptors, the model achieved an overall MSE of 0.0105 and a correlation coefficient of 0.9306, and the model revealed that the synergistic interplay of molecular polarity, lipophilicity, and structural complexity predominantly governs variations in both HTC and CHF. Subsequently, COSMO descriptors were introduced to provide a more detailed, fragment-level representation of molecular surface charge. The COSMO-based model achieved superior predictive performance, with an overall mean squared error of 0.0094 and correlation coefficient $R = 0.9797$. In the feature-importance analysis, charge density accounted for 86 % of the variance in HTC and CHF. Sensitivity analysis further revealed that the S4 (non-polar) region in the σ-profile had a sensitivity parameter of 2.05 for HTC but −0.893 for CHF, whereas the S5 (HBA) region exhibited a parameter of 0.647 for CHF yet −0.868 for HTC. Mechanistic interpretation indicates that non-polar and HBD-rich regions disrupt interfacial water structure, lowering the energetic cost of vapor embryo formation, increasing nucleation density, and accelerating bubble birth-departure cycles, thereby enhancing HTC. Conversely, HBA-rich regions anchor liquid via directional hydrogen bonding and enhanced polar interactions, stabilizing the thin liquid film and improving capillary- and molecular-level rewetting under high heat flux, thus raising CHF. This study bridges predictive machine-learning modeling with mechanistic insight by proposing a descriptor-property-mechanism framework for rational surfactant design. The elucidated roles of charge density and interfacial regions provide clear guidelines for tailoring molecules to optimize HTC and CHF. Future work may apply high-throughput screening and multi-scale validation to accelerate the development of advanced surfactants for thermal management and fire protection.

CRediT authorship contribution statement

Hao Xiao: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Tao Yang:** Resources, Formal analysis, Data curation. **Yi Zheng:** Validation, Supervision, Resources. **Guangwen Zhang:** Supervision, Formal analysis. **Ming Ma:** Formal analysis, Data curation. **Chunyu Jiang:** Software, Resources. **Chun Wang:** Project administration, Methodology. **Xiangdi Zhao:** Supervision, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] B. Hanqi, G. Xia, H. Ma, W. Zhou, Facile fabrication of ag nanoparticles-coated surfaces for enhanced pool boiling heat transfer, Int. Commun. Heat Mass Transf. 169 (2025) 109539.

[2] S. Huang, G. Zhang, Y. Li, X. Luo, R. Yuan, Simulation and experiment of boiling two-phase flow and heat transfer in mini-channels with reticulated pore structure, Int. Commun. Heat Mass Transf. 164 (2025) 108899.
[3] Y. Wada, Y. Sibamoto, T. Hibiki, Heat transfer characteristics of downward saturated boiling flow in vertical round pipes, Int. J. Heat Mass Transf. 239 (2025) 126598.
[4] Q. Wang, H. Ren, P. Huang, D.-C. Gao, Y. Sun, Multiscale hybrid surface structure modifications for enhanced pool boiling heat transfer: state-of-the-art review, Renew. Sust. Energ. Rev. 208 (2025) 115018.
[5] M.R. Ali, K. Al-Khaled, M. Hussain, T. Labidi, S.U. Khan, L. Kolsi, R. Sadat, Effect of design parameters on passive control of heat transfer enhancement phenomenon in heat exchangers–a brief review, Case Stud. Therm. Eng. 43 (2023) 102674.
[6] S. Lei, C. Hu, Z. Li, S. Tan, C. Wang, Y. Wu, Boiling heat transfer enhancement on heterogeneous copper foams in the presence of transverse flow, Appl. Therm. Eng. 260 (2025) 124990.
[7] J. Pereira, R. Souza, R. Lima, A. Moreira, A. Moita, An overview of the recent advances in Pool boiling enhancement materials, Structrure, and devices, Micromachines 15 (2) (2024) 281.
[8] Z. Xu, C. Ding, X. Wang, Z. Zhang, J. Xu, Y. Qiu, D. Shan, B. Guo, Bottom-up design of micro−/nano hierarchical surface for the enhancement of boiling heat transfer performance: a review, Int. J. Heat Mass Transf. 234 (2024) 126131.
[9] C. Zhu, L. Zhang, D. Zhang, Z. Wang, L. Qian, Z. Jiang, Effect of micro-cavities structured surfaces on bubble dynamics and pool boiling heat transfer enhancement, Appl. Therm. Eng. 270 (2025) 126220.
[10] M. Bongarala, J.A. Weibel, S.V. Garimella, Boiling crisis is a consequence of thermal runaway in the substrate due to degradation of nucleate boiling heat transfer, Int. J. Heat Mass Transf. 233 (2024) 125906.
[11] W. Liu, Z. Liang, Y. Luo, Enhanced boiling heat transfer performance on wettability-patterned surface, Appl. Therm. Eng. 244 (2024) 122792.
[12] G. Liang, I. Mudawar, Review of pool boiling enhancement with additives and nanofluids, Int. J. Heat Mass Transf. 124 (2018) 423–453.
[13] B. Ravikumar, I.K. Karathanassis, T. Smith, M. Gavaises, Atomistic investigation of viscoelastic nanofluids as heat transfer liquids for immersive-cooling applications, Ind. Eng. Chem. Res. 63 (48) (2024) 21023–21037.
[14] S. Ni, C.K. Cheng, X. Li, S. Ahmad, D. Ji, J. Zhao, The hydrodynamic aspect mechanism of the effect of surfactant on boiling heat transfer, Int. Commun. Heat Mass Transf. 156 (2024) 107694.
[15] S. Li, S.U. Khan, K. Al-Khaled, E. Ali, M.I. Khan, K.M.A. Elamin, B.M. Fadhl, B. M. Makhdoum, Development of convective heat transport in nanofluid flow of Oldroyd-B model with magnetic dipole moment, Z. Angew. Math. Me. 104 (1) (2024) 202200564.
[16] S.K. Singh, D. Sharma, Review of pool and flow boiling heat transfer enhancement through surface modification, Int. J. Heat Mass Transf. 181 (2021) 122020.
[17] H. Yang, A. Majeed, K. Al-Khaled, T. Abbas, M. Naeem, S.U. Khan, M. Saeed, Significance of melting heat transfer and Brownian Motion on flow of Powell–Eyring fluid conveying nano-sized particles with improved energy systems, Lubricants 11 (1) (2023) 32.
[18] V.K. Kandasamy, S. Rajendran, S.J.D. Vijayakumar, S. Paul Singarayar, Nanofluid nucleate boiling assessment on heating surfaces: a comprehensive study, J. Therm. Anal. Calorim. 148 (15) (2023) 7687–7705.
[19] A. Hadžić, M. Može, K. Arhar, M. Zupančič, I. Golobič, Effect of nanoparticle size and concentration on pool boiling heat transfer with TiO_2 nanofluids on laser-textured copper surfaces, Nanomaterial 12 (15) (2022) 2611.
[20] H. Jia, L. Xu, X. Xiao, K. Zhong, Study on boiling heat transfer of surfactant solution on grooved surface, Int. J. Heat Mass Transf. 181 (2021) 121876.
[21] T. Wen, J. Luo, K. Jiao, L. Lu, Comparison of pool boiling heat transfer performance between aqueous cationic and anionic surfactant solutions with similar ionic group, Appl. Therm. Eng. 216 (2022) 119136.
[22] J. Luo, H. Yang, A novel surfactant material for performance enhancement on nucleate pool boiling heat transfer, Int. Commun. Heat Mass Transf. 144 (2023) 106800.
[23] J. Luo, H. Yang, T. Wen, Heat transfer enhancement and long-term test of non-ionic Triton surfactant with different hydrophilic chain lengths, Int J Therm Sci 200 (2024) 108967.
[24] G. Liang, I. Mudawar, Pool boiling critical heat flux (CHF) – part 2: assessment of models and correlations, Int. J. Heat Mass Transf. 117 (2018) 1368–1383.
[25] H. Ahmad, K. Al-Khaled, A.S. Sowayan, M. Abdullah, M. Hussain, A. Hammad, S. U. Khan, I. Tlili, Experimental investigation for automotive radiator heat transfer performance with $ZnO–Al_2O_3$/water-based hybird nanoparticles: an improved thermal model, Int. J. Mod. Phys. B 37 (05) (2023) 2350050.
[26] T. Kinjo, Y. Sei, N. Giannetti, K. Saito, K. Enoki, Prediction of boiling heat transfer coefficient for micro-fin using mini-channel, Appl. Sci. 14 (15) (2024) 6777.
[27] S.-J. Li, J. Li, X.-B. Zhang, Z.-H. Luo, Analysis of spatial nonuniform behaviors of subcooled flow boiling employing Euler–Lagrange approach, Ind. Eng. Chem. Res. 63 (31) (2024) 13917–13928.
[28] S. Alipour Bonab, M. Yazdani-Asrami, Artificial intelligence-based model to predict the heat transfer coefficient in flow boiling of liquid hydrogen as fuel and cryogenic coolant in future hydrogen-powered cryo-electric aviation, Fuel 381 (2025) 133323.
[29] J.R. Kitchin, Machine learning in catalysis, Nat. Catal. 1 (4) (2018) 230–232.
[30] D. Fan, K. Xue, R. Zhang, W. Zhu, H. Zhang, J. Qi, Z. Zhu, Y. Wang, P. Cui, Application of interpretable machine learning models to improve the prediction performance of ionic liquids toxicity, Sci. Total Environ. 908 (2024) 168168.
[31] R. Zhang, Y. Wang, W. Zhu, L. Xin, J. Qi, Y. Wang, Z. Zhu, P. Cui, Insight into the mechanism of machine learning models for predicting ionic liquids toxicity based on molecular structure descriptors, ACS Sustain. Chem. Eng. 12 (49) (2024) 17749–17760.
[32] I.T. Sung, Y.-H. Cheng, C.-M. Hsieh, L.-C. Lin, Machine learning for gas adsorption in metal–organic frameworks: a review on predictive descriptors, Ind. Eng. Chem. Res. 64 (4) (2025) 1859–1875.
[33] M.A. Javed, A. Ghaffari, H.M. Atif, A.S. Sowayan, S.U. Khan, Machine learning non-isothermal study of the blade coating process (NIS-BCP) using non-Newtonian nanofluid with magnetohydrodynamic (MHD) and slip effects, Polym. Compos. 32 (2024), 09673911241282424.
[34] A. Upadhyay, B. Kumar, R. Raj, Ionic liquid as a cosurfactant for critical heat flux enhancement during boiling with aqueous surfactant solutions, Appl. Therm. Eng. 246 (2024) 122962.
[35] A. Upadhyay, B. Kumar, N. Kumar, R. Raj, Simultaneous enhancement of critical heat flux and heat transfer coefficient via in-situ deposition of ionic liquids during pool boiling, Int. J. Heat Mass Transf. 208 (2023) 124066.
[36] R.K. Gouda, M. Pathak, M. Kaleem Khan, A biosurfactant as prospective additive for pool boiling heat transfer enhancement, Int. J. Heat Mass Transf. 150 (2020) 119292.
[37] T. Wen, J. Luo, K. Jiao, L. Lu, Pool boiling heat transfer enhancement of aqueous solution with quaternary ammonium cationic surfactants on copper surface, Int. J. Heat Mass Transf. 190 (2022) 122761.
[38] A. Klamt, The COSMO and COSMO-RS solvation models, WIREs Comput. Mol. Sci. 8 (1) (2018) 1338.
[39] C. Gui, M. Song, Q. Liu, D. Xiang, P. Lu, S. Fourmentin, C. Hua, Z. Lei, Rational screening of deep eutectic solvents for the removal of halogenated volatile organic compounds, AICHE J. 71 (8) (2025) 18858.
[40] Z. Chen, H. Zhang, H. Li, Y. Xu, Y. Shen, Z. Zhu, J. Gao, Y. Ma, Y. Wang, Separation of n-heptane and tert-butanol by ionic liquids based on COSMO-SAC model, Green Energy Environ. 6 (3) (2021) 380–391.